

Matrices: Multiplication

Definition

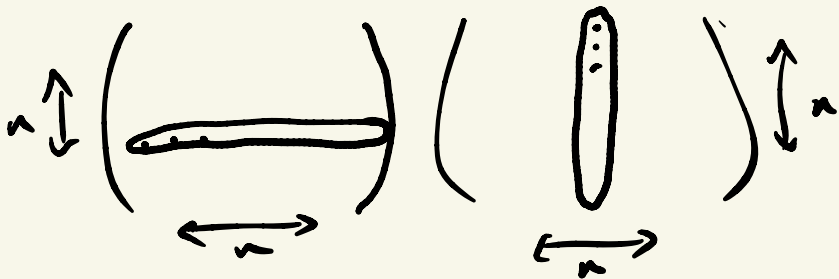
Suppose that A is an $m \times k$ matrix and $B = [b_1 \ b_2 \ \dots \ b_n]$ is a $k \times n$ matrix then we define the product AB by

$$[Ab_1 \ Ab_2 \ \dots \ Ab_n]$$

which is an $m \times n$ matrix.

Matrix multiplication as composition

If A is $m \times k$ and B is $k \times n$ then AB is the standard matrix of the composition of the matrix transformations $T_A \circ T_B$.



Two $n \times n$ matrices. How hard is it to multiply them from a time point of view?

To get any entry in the product we need to do n multiplications and then $n-1$ additions $\approx 2n$ operations

To get the whole multiplication we need
 $\approx 2n^3$ operations.

$\uparrow$
order n^3

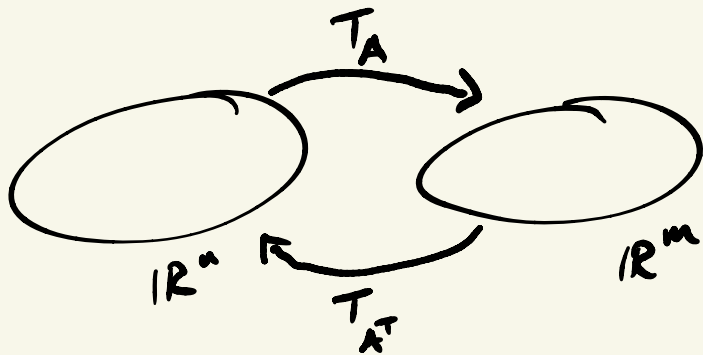
Matrices: Powers and the transpose

- If A is an $n \times n$ matrix then for an k we can multiply A together k times and write A^k . We say $A^0 = I_n$.
- If $A = (a_{ij})$ is $m \times n$ then the transpose of A , written A^T , is the $n \times m$ matrix where the ij entry is a_{ji} i.e. the rows of A become the columns of A^T and vice versa.

For all matrices A and B of the correct size and scalars λ

- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T \leftarrow !$
- $(\lambda A)^T = \lambda A^T$

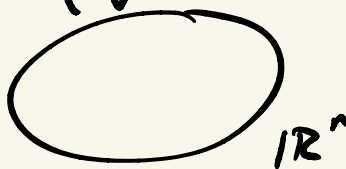
A is $m \times n$, $T_A: \mathbb{R}^n \rightarrow \mathbb{R}^m$



$$\underline{(AB)^T = B^T A^T}$$

A is $n \times n$

$T_A \in \text{linear operator.}$



Invertible Matrices

Definition

We say an $n \times n$ matrix A is invertible if there is a matrix B such that $AB = BA = I_n$. We call B the inverse of A and write A^{-1} for B .

Fact:

- If A is invertible then A^{-1} is uniquely defined.
- If A is invertible then the associated linear transformation is one-to-one and onto.

$$I_n = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \leftarrow n \times n \text{ identity matrix.}$$

Ex: $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ $B = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$
 $\uparrow$ B the inverse

Check: $AB = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$

$BA = (\text{check}) = I_2.$

So B is the inverse of A .

Let's check that if A has two inverses
B and C then in fact $B = C$.

Compute BAC two ways:

$$BAC = (BA)C = I_n C = C \quad \text{because } BA = I_n.$$

$$BAC = B(AC) = BI_n = B \quad \text{because } AC = I_n.$$

$$\text{So } B = C!$$

If A is invertible then T_A is both 1-1
and onto.

1) Suppose A is invertible and $Ax = 0$

$$\underline{A}^{-1}(\underline{Ax}) = \underline{A}^{-1}0 = 0$$

so $I_n x = 0$ i.e. $x = 0$ so T_A is 1-1.

2) To see that T_A is onto, pick $b \in \mathbb{R}^n$.

We need x s.t. $Ax = b$. Let $x = A^{-1}b$.

$$\text{and then } A(A^{-1}b) = (AA^{-1})b = b.$$

A little bit of algebra with inverses

- If A is invertible then so is A^{-1} and $(A^{-1})^{-1} = A$.
- If A and B are invertible and of the same size then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.
- If A is $n \times n$ and invertible then so is A^T and $(A^T)^{-1} = (A^{-1})^T$.

Claim: If A is invertible then so is A^{-1} and $(A^{-1})^{-1} = A$.

We need to show that A is the inverse of A^{-1} .

$$\underline{A}A^{-1} = I_n \quad A^{-1}A = I_n.$$

$$\text{So } (A^{-1})^{-1} = A!$$

Show $(AB)^{-1} = \underline{B^{-1}A^{-1}}$ assuming all matrices
sense.

Check: $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1}$
 $= AI_n A^{-1}$
 $= AA^{-1} = I_n.$

$$\begin{aligned}(B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B \\ &= B^{-1}I_n B \\ &= B^{-1}B = I_n.\end{aligned}$$